

Face detection

1. First determine whether the version of the image transmission module is the AI version firmware. **If you already know the firmware version, you can ignore this step** and use the **wifi_ver** command to query, as shown in the figure below

```
[2024-08-06 14:53:29.948]# SEND ASCII>
wifi_ver
[2024-08-06 14:53:30.010]# RECV ASCII>
YAHBOOM VerSion:AI_V0.5.0
```

2. Switch the mode to face detection mode. Use the following command to send **ai_mode:2** through the serial assistant. After successful configuration, the module will restart and print the current firmware version. Then, face detection can start, as shown in the figure

```
[2024-08-06 15:03:04.205]# SEND ASCII>
ai_mode:2
[2024-08-06 15:03:04.341]# RECV ASCII>
ESP-ROM:esp32s3-20210327
Build:Mar 27 2021
rst:0x3 (RTC_SW_SYS_RST),boot:0x8 (SPI_FAST_FLASH_BOOT)
Saved PC:0x40375a0c
SPIWP:0xee
mode:DIO, clock div:1
load:0x3fce3818,len:0x514
load:0x403c9700,len:0x4
load:0x403c9704,len:0xb58
load:0x403cc700,len:0x2a8c
entry 0x403c98fc

[2024-08-06 15:03:04.886]# RECV ASCII>
[0:32mI (549) cpu_start: Multicore app[0m
[0:32mI (549) esp_psram: Found 2MB PSRAM device[0m
[0:32mI (550) esp_psram: Speed: 80MHz[0m
[0:32mI (550) cpu_start: Pro cpu up. [0m
[0:32mI (554) cpu_start: Starting app cpu, €
[2024-08-06 15:03:07.070]# RECV ASCII>
YAHBOOM AI VerSion:AI_V0.5.0
```

Note: If the current mode is already face detection mode, the module only returns ok

```
[2024-08-06 15:05:32.444]# SEND ASCII>
ai_mode:2
[2024-08-06 15:05:32.503]# RECV ASCII>
OK
```

3. Start recognizing faces If recognized, the coordinates of the upper left corner and lower right corner of the first frame will be sent out through the serial port.

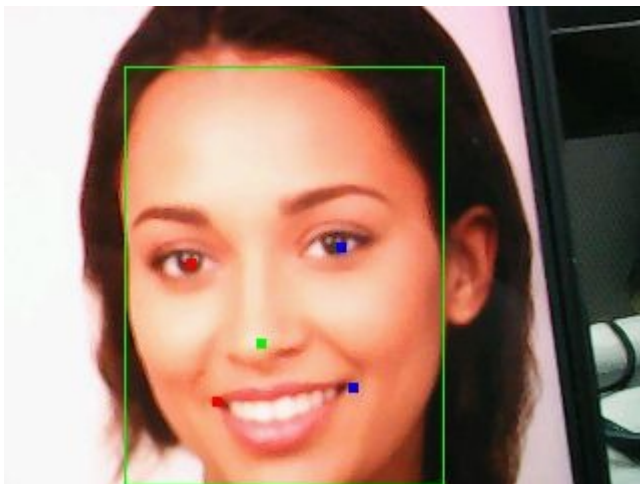
The specific content is **\$upper left corner x-axis coordinate, upper left corner y-axis coordinate, lower right corner x-axis coordinate, lower right corner y-axis coordinate, #**

Note: You need to open the screen to start recognition. If the screen is closed in the middle, the recognition will also be closed.

Face



Result:



Serial port information:

\$51,40,204,187,#	\$44,36,198,191,#	\$40,43,196,191,#
\$70,48,212,184,#	\$40,41,198,192,#	\$58,40,213,186,#
\$76,40,211,179,#	\$63,32,201,178,#	\$68,28,210,183,#
\$71,28,218,185,#	\$72,32,220,182,#	\$80,35,220,174,#
\$41,80,185,237,#	\$50,97,188,222,#	\$50,116,182,221,#
\$35,99,194,241,#	\$49,112,187,221,#	\$55,77,194,212,#
\$52,73,217,222,#	\$42,65,202,210,#	\$38,60,203,211,#
\$42,67,198,207,#	\$39,61,205,209,#	\$59,89,204,205,#
\$40,67,193,209,#	\$35,61,198,213,#	\$37,55,209,210,#
\$39,59,194,207,#	\$37,58,200,208,#	\$40,59,194,208,#
\$41,67,193,207,#	\$36,63,195,212,#	\$31,66,191,212,#
\$42,62,192,204,#	\$45,64,189,203,#	\$67,80,209,207,#
\$42,63,185,203,#	\$43,66,186,203,#	\$57,70,194,201,#
\$47,65,188,199,#	\$45,66,184,201,#	\$45,64,188,202,#
\$56,71,194,199,#	\$60,76,194,201,#	\$42,62,192,204,#
\$56,78,194,200,#	\$78,82,216,212,#	\$40,62,191,206,#
\$41,61,189,205,#	\$54,70,199,204,#	\$53,65,196,199,#
\$56,77,205,206,#	\$58,66,195,200,#	\$43,62,191,201,#
\$70,84,205,191,#	\$37,57,185,200,#	\$38,50,191,202,#
\$58,77,204,199,#	\$29,43,210,219,#	\$37,55,186,202,#
\$33,41,212,206,#	\$31,43,209,218,#	\$32,49,191,204,#
\$34,48,193,202,#	\$32,46,194,202,#	\$29,37,203,205,#
\$25,42,204,198,#	\$28,37,201,200,#	\$42,51,194,193,#
\$23,40,204,202,#	\$71,72,206,188,#	\$31,42,199,186,#
\$31,39,203,189,#	\$29,46,203,185,#	\$38,49,193,194,#
\$33,43,203,179,#	\$33,41,209,178,#	\$42,37,204,176,#
\$36,34,219,181,#	\$36,34,218,180,#	\$39,37,209,177,#